

Modeling GLP-1 Receptor Agonists for Weight Loss

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Abstract— G protein coupled receptors (GPCRs) represent pivotal targets for medications aimed at targeting a range of chronic conditions such as obesity and diabetes[1]. Obesity, a multifaceted chronic ailment, significantly elevates the likelihood of developing severe conditions such as type 2 diabetes, heart disease and certain cancers[1]. Hence, it is imperative to prioritize obesity management to avert the onset of more life-threatening illnesses. Since the advent of the initial GLP-1 receptor agonist (GLP-1RA) model, patients have seen remarkable achievements, including double-digit weight loss and cardiovascular advantages[1]. The triumph of this pioneering GLP-1 approach has unveiled fresh understandings regarding molecular factors influencing GLP-1RA ligand selectivity and functional efficacy[2]. These insights have opened up avenues for crafting and refining more potent remedies for metabolic conditions, such as an orally administered tablet formulation [1], [2].

Keywords— GLP-1RA, GPCRs, Obesity

I. INTRODUCTION

Obesity is a chronic disease defined by excessive fat deposits that can impair health. It can also lead to type II diabetes, heart disease, and affect bone health and reproduction. Causes of obesity include; lack of physical activity, unhealthy eating behaviours, environmental causes or genetics.

G-protein coupled receptors (GPCRs) are the most extensive group of membrane proteins that are responsible for regulating many important biological processes. The dysregulation of GPCRs can be used to target various chronic diseases, including diabetes and obesity [3]. Specifically, GLP-1 is a peptide hormone secreted by the gut L cells that improve glucose homeostasis and promote weight loss. GLP-1 stimulates the release of insulin

from pancreatic beta cells in response to elevated blood sugar levels [3]. This helps to lower the glucose levels after meals. When insulin levels are high, GLP-1 influences ATP-sensitive or voltage-gated potassium channels. This causes membrane depolarization and the release of calcium ions from both internal and external storage. The increase in calcium along with elevated cAMP levels stimulate the release of insulin. GLP-1 also controls appetite by promoting the feeling of “fullness” by acting on the part of the brain for appetite control [4]. Due to its beneficial effects on blood sugar and weight management, synthetic versions of GLP-1, known as GLP-1 receptor agonists, have been developed to treat obesity [4].

II. MECHANISM OF ACTION

GLP-1 receptor agonists have a broad impact on the weight control system in the human body. The action first starts with the interaction between GLP-1 RA and the central nervous system, which plays a role in central appetite and satiety regulation. Further, the action of GLP-1 RA then goes down to the cellular level, leading to the activation of specific signal transduction pathways called G protein-coupled receptor pathways. These pathways then play a leading role in determining weight reduction based on the impact of these pathways on various cellular events related to metabolic and energy processes.

A. Physiologic Model

In healthy individuals, when food is consumed, our gastrointestinal tract releases hormones such as GLP-1 and GIP [5]. The pancreas then releases insulin and amylin in response [5]. In synergy, amylin and GLP-1 have inhibitory effects on gastric emptying, glucagon release and appetite, which is illustrated in figure 1 [5]. The adsorption of GLP-1 and GIP result in increased insulin secretion. This is known as the incretin effect [5]. The human body

uses the incretin effect to maintain the balance of blood sugar levels, or glucose homeostasis [6]. Obese individuals have this balance disrupted. GLP-1 receptor agonists are medications that act as the natural GLP-1 hormone but at higher concentrations [6]. These drugs have an increased half life than the body's own GLP-1, thereby promoting sustained incretin activity [6]. This long-lasting effect of the synthetic hormone enables long-term weight management in obese patients.

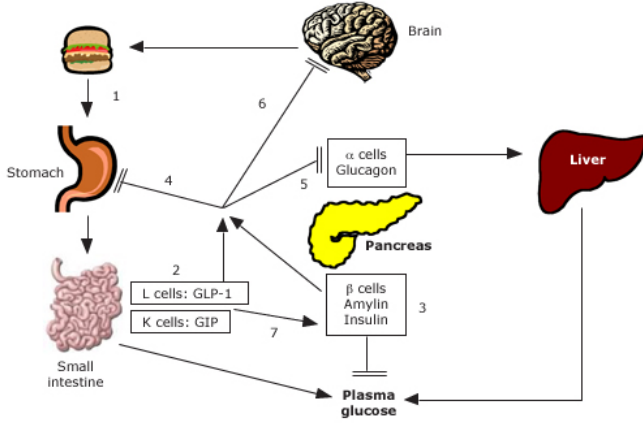


Fig 1: A model of the mechanism of action of the incretin effect in the human body [5].

B. Intracellular Model

The GLP-1RA protein is a class B GPCR whose receptors include hormones such as glucagon, secretin, and GIP. The receptors are found on the membrane of pancreatic β cells [2]. As illustrated in figure 2, when GLP-1 binds to a receptor, it elicits adenylyl cyclase (AC) activity [2]. The activation of AC along with ATP, results in cAMP formation. This then leads to the activation of protein kinase A (PKA) which elicits a cellular response [2]. In addition to this, in the presence of high glucose levels, GLP-1 also has an effect on voltage-gated potassium and calcium channels [2]. Because these channels are sensitive to ATP and the GLP-1 pathway is dependent on ATP, the GLP-1 pathway has inhibitory effects on these voltage gated channels [2].

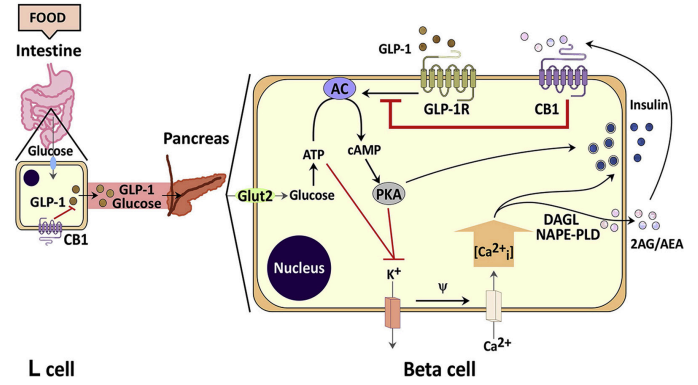


Fig 2: A model of the GLP-1 intracellular mechanism of action

III. PROTEIN MODELING

The GLP-1 receptor is primarily found on the surface of pancreatic β cells. When the GLP-ligand binds to the pancreatic β receptor, it triggers a series of intracellular signaling events that lead to increased insulin secretion, among other effects. GLP-1 binding to its receptor activates protein Heterotrimeric Gs, which then increases the activity of adenylyl cyclase. Activated adenylyl cyclase catalyzes the conversion of ATP to cAMP. Elevated levels of cAMP then activates protein kinase A (PKA), which phosphorylates various target proteins in insulin secretion. This leads to an increased calcium level, along with cAMP, which stimulates the release of insulin [4].

A. Network Topology

The network topology illustrates the cascading effect the ligand binding to the receptor has which is illustrated by figure 3. When the ligand binds to the receptor it creates the ligand receptor complex. This triggers activity of AC and ATP which then produces cAMP. The cAMP is then able to activate PKA, which then produces a cellular response. All of these processes are reversible. This network topology allowed us to model the GLP-1 pathway using a series of differential equations.

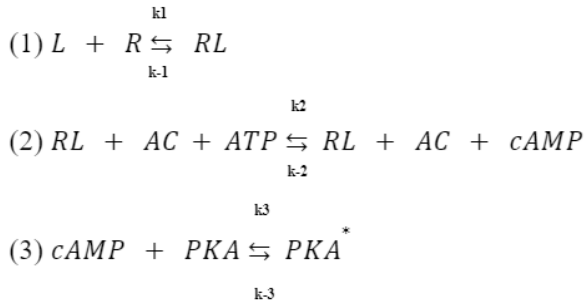


Fig 3: A model of the intracellular network topology of the GLP-1 pathway

Species	Name
L	GLP-1 Ligand
R	Pancreatic β Cell Receptor
LR	Ligand-Receptor Complex
AC	Adenylyl Cyclase
ATP	Adenosine Triphosphate
cAMP	Cyclic AMP
PKA	Protein Kinase A
PKA*	Activated Protein Kinase A

Fig 4: A key for the species used in the network topology

B. Differential Equations

$$\begin{aligned}
(4) \quad & \frac{d}{dt} [R](t) = k_{-1}[RL](t) - k_1[R](t)[L](t) \\
(5) \quad & \frac{d}{dt} [L](t) = k_{-1}[RL](t) - k_1[R](t)[L](t) \\
(6) \quad & \frac{d}{dt} [RL](t) = k_1[R](t)[L](t) - k_{-1}[RL](t) - k_2[RL](t)[AC](t)[ATP](t) \\
& \quad + k_{-2}[RL](t)[AC](t)[cAMP](t) \\
(7) \quad & \frac{d}{dt} [AC](t) = k_{-2}[RL](t)[AC](t)[cAMP](t) - k_2[RL](t)[AC](t)[ATP](t) \\
(8) \quad & \frac{d}{dt} [ATP](t) = k_{-2}[RL](t)[AC](t)[cAMP](t) - k_2[ATP](t)[AC](t)[RL](t) \\
(9) \quad & \frac{d}{dt} [PKA](t) = k_{-3}[PKA^*] - k_3[cAMP](t)[PKA](t) \\
(10) \quad & \frac{d}{dt} [cAMP](t) = k_2[RL](t)[AC](t)[ATP](t) - k_{-2}[RL](t)[AC](t)[cAMP](t) \\
& \quad - k_3[cAMP](t)[PKA](t) + k_{-3}[PKA^*] \\
(11) \quad & \frac{d}{dt} [PKA^*](t) = k_3[cAMP](t)[PKA](t) - k_{-3}[PKA^*]
\end{aligned}$$

Fig 5: A model of the mechanism of action of the incretin effect in the human body

Differential equations for each concentration was constructed using the general form:

$$\frac{d}{dt} R(t) = \text{rate of production} - \text{rate of consumption}$$

and was repeated for every concentration variable. It is important to note that in equation 2, both RL and AC appear twice, in the reactant and product side of the equation. An assumption was made that the concentration of RL and AC are greater on the reactant side in order to simplify the differential equations.

Equations 4 and 5 represent the rate at which the receptor-ligand complex RL dissociates back into the receptor R and ligand L respectively. The rate constants and the initial concentrations were chosen at random. Equation 7 represents the balance between the activated receptor-ligand complex RL and the inhibition by ATP. These equations collectively model the dynamics of a biochemical signaling pathway involving interactions between receptors, ligands, enzymes, and second messengers.

IV. RESULTS

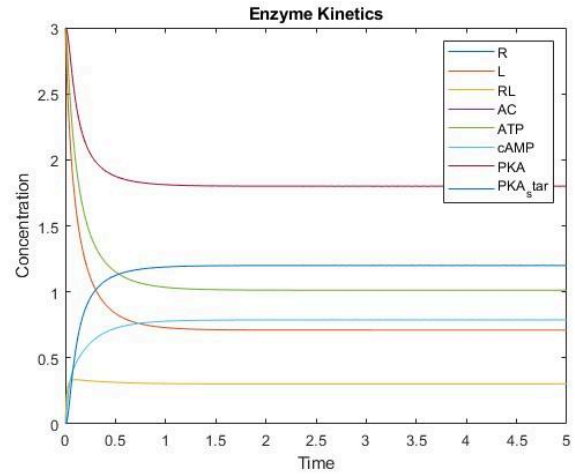


Figure 6: A graphical representation of the Enzyme Kinetics of each variable concentration

Using MATLAB, the differential equations were solved and graphed using the ode45 solver. The entire code is included in the Appendix.

V. CONCLUSIONS

Based on the graph given the differential equations and the assumed variables, a few conclusions can be drawn. Firstly, it is evident that the concentration of RL does not change over time, or is in steady state. Secondly, the concentrations of PKA, ATP, and L decrease over time and are in a transient state. Thirdly, the concentrations of R and cAMP increase over time and are in a transient state.

VI. FUTURE WORK

Future opportunities may arise from: (i) deorphanization of orphan GPCRs to provide novel targets; (ii) new indication for drug intervention via discovery and/or repurposing efforts; (iii) development of lead compounds targeting classes B2 and F GPCRs to address unmet medical needs; and (iv) validation of polypharmacology may lead to improved drug therapies

artificial intelligence integration new scaffolds
wait for first orally active gpcr-based incretin secretive medicine

class A gpcr family challenge structure image
hinder rational design potent

selectivity target multiple gpcr
find right balance optimal ratio of each gpcr
against entity in a unimolecular poly-agonist

drug screening tools
pharmacodynamic pharmacokinetic
considerations inhibitor moiety present distinct
receptor levels turnover rates desensitized
mechanisms psychophysiological

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